Week 8: Database Relationship Analysis Using Joins

Organic Grocery Store Management System

Find 1: Display customer information with their orders.

```
SQL> SELECT c.First_Name, c.Last_Name,o.Order_ID, o.Order_Date, o.Total_Amou
nt, o.Order_Status FROM Customer c INNER JOIN Orders o ON c.Customer_ID = o.
Customer_ID;

FIRST_NAME
--------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT ORDER_STATUS
------------ --------------------
Arun
Kumar                                                   101 01-SEP-26
         500 Delivered

Priya
Raj                                                     102 02-SEP-26
         750 Pending

FIRST_NAME
--------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT ORDER_STATUS
------------ --------------------

Arun
Kumar                                                   103 03-SEP-26
         300 Shipped

Kavin
M                                                       104 04-SEP-26

FIRST_NAME
--------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT ORDER_STATUS
------------ --------------------
         900 Delivered
```

```
FIRST_NAME
--------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT ORDER_STATUS
------------ --------------------

Arun
Kumar                                                 103 03-SEP-26
         300 Shipped

Kavin
M                                                     104 04-SEP-26

FIRST_NAME
--------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT ORDER_STATUS
------------ --------------------
         900 Delivered

Priya
Raj                                                   105 05-SEP-26
         400 Pending
```

Find 2: Display order details with payment information

```
SQL> SELECT od.Order_ID, od.Product_ID,od.Quantity, p.Payment_Mode,p.Payment
_Status, p.Transaction_Amount FROM Order_Details od INNER JOIN Payment p ON
od.Order_ID = p.Order_ID;

no rows selected
```

Find 3: Retrieve products purchased by customers

```
SQL> SELECT c.First_Name, p.Product_Name,od.Quantity, od.Price FROM Customer
 c INNER JOIN Orders o ON c.Customer_ID = o.Customer_ID INNER JOIN Order_Det
ails od ON o.Order_ID = od.Order_ID INNER JOIN Product p ON od.Product_ID =
p.Product_ID;

no rows selected
```

2. Implement LEFT JOIN

Find 1: Display all customers, including customers without orders

```
SQL> SELECT c.First_Name, c.Last_Name, o.Order_ID FROM Customer c LEFT JOIN
Orders o ON c.Customer_ID = o.Customer_ID;

FIRST_NAME
--------------------------------------------------
LAST_NAME                                          ORDER_ID
-------------------------------------------------- ----------
Arun
Kumar                                                     101

Priya
Raj                                                       102

Arun
Kumar                                                     103


FIRST_NAME
--------------------------------------------------
LAST_NAME                                          ORDER_ID
-------------------------------------------------- ----------
Kavin
M                                                         104

Priya
Raj                                                       105
```

Find 2: Find customers without any purchases

```
SQL> SELECT c.First_Name, c.Last_Name FROM Customer c LEFT JOIN Orders o ON
c.Customer_ID = o.Customer_ID WHERE o.Order_ID IS NULL;

no rows selected
```

Find 3: Display all products, including products with no sales

```
SQL> SELECT c.First_Name, c.Last_Name FROM Customer c LEFT JOIN Orders o ON
c.Customer_ID = o.Customer_ID WHERE o.Order_ID IS NULL;

no rows selected

SQL> SELECT p.Product_Name, od.Order_ID, od.Quantity FROM Product p LEFT JOI
N Order_Details od ON p.Product_ID = od.Product_ID;

PRODUCT_NAME
--------------------------------------------------------------------------
----
  ORDER_ID   QUANTITY
---------- ----------
Organic Apple


Organic Banana


Organic Carrot



PRODUCT_NAME
--------------------------------------------------------------------------
----
  ORDER_ID   QUANTITY
---------- ----------
Brown Rice


Green Tea
```

3. Implement RIGHT JOIN

Find 1: Display all orders with customer information

```
SQL> SELECT c.First_Name, c.Last_Name,o.Order_ID, o.Order_Date FROM Customer
 c RIGHT JOIN Orders o ON c.Customer_ID = o.Customer_ID;

FIRST_NAME
--------------------------------------------------
LAST_NAME                                            ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
Arun
Kumar                                                     103 03-SEP-26

Arun
Kumar                                                     101 01-SEP-26

Priya
Raj                                                       105 05-SEP-26


FIRST_NAME
--------------------------------------------------
LAST_NAME                                            ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
Priya
Raj                                                       102 02-SEP-26

Kavin
M                                                         104 04-SEP-26
```

Find 2: Find orders where customer details are missing

```
SQL> SELECT o.Order_ID, o.Customer_ID FROM Customer c RIGHT JOIN Orders o ON
 c.Customer_ID = o.Customer_ID WHERE c.Customer_ID IS NULL;

no rows selected
```

Find 3: Display all payment records with order details

```
SQL> SELECT p.Payment_ID, p.Order_ID, p.Payment_Mode, p.Payment_Status,od.Pr
oduct_ID, od.Quantity FROM Order_Details od RIGHT JOIN Payment p ON od.Order
_ID = p.Order_ID;

PAYMENT_ID   ORDER_ID PAYMENT_MODE         PAYMENT_ST PRODUCT_ID   QUANTITY
---------- ---------- -------------------- ---------- ---------- ----------
         3        103 Cash                 Success
         2        102 Card                 Failed
         1        101 UPI                  Success
         5        105 UPI                  Success
         4        104 Net Banking          Pending
```

4. Retrieve Complete Order Details

```
SQL> SELECT c.First_Name, c.Last_Name,pr.Product_Name, od.Quantity,o.Order_D
ate, o.Total_Amount,p.Payment_Status FROM Customer c JOIN Orders o ON c.Cust
omer_ID = o.Customer_ID JOIN Order_Details od ON o.Order_ID = od.Order_ID JO
IN Product pr ON od.Product_ID = pr.Product_ID LEFT JOIN Payment p ON o.Orde
r_ID = p.Order_ID;

no rows selected
```

5. Display Customer Purchase History

Find 1: Total amount spent by each customer

```
SQL> SELECT c.First_Name, c.Last_Name,NVL(SUM(o.Total_Amount), 0) AS Total_S
pent FROM Customer c LEFT JOIN Orders o ON c.Customer_ID = o.Customer_ID GRO
UP BY c.Customer_ID, c.First_Name, c.Last_Name;

FIRST_NAME
--------------------------------------------------
LAST_NAME                                          TOTAL_SPENT
-------------------------------------------------- -----------
Arun
Kumar                                                      800

Kavin
M                                                          900

Priya
Raj                                                       1150
```

Find 3: Number of orders placed by each customer

```
SQL> SELECT c.First_Name, c.Last_Name,COUNT(o.Order_ID) AS Total_Orders FROM
 Customer c LEFT JOIN Orders o ON c.Customer_ID = o.Customer_ID GROUP BY c.C
ustomer_ID, c.First_Name, c.Last_Name;

FIRST_NAME
--------------------------------------------------
LAST_NAME                                          TOTAL_ORDERS
-------------------------------------------------- ------------
Arun
Kumar                                                         2

Kavin
M                                                             1

Priya
Raj                                                           2
```

Find 4: Latest purchase details of customers

```
SQL> SELECT c.First_Name, c.Last_Name,o.Order_ID, o.Order_Date,o.Total_Amoun
t FROM Customer c JOIN Orders o ON c.Customer_ID = o.Customer_ID WHERE o.Ord
er_Date = (SELECT MAX(o2.Order_Date)FROM Orders o2 WHERE o2.Customer_ID = c.
Customer_ID );

FIRST_NAME
--------------------------------------------------
LAST_NAME                                            ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------
Arun
Kumar                                                     103 03-SEP-26
         300

Priya
Raj                                                       105 05-SEP-26
         400

FIRST_NAME
--------------------------------------------------
LAST_NAME                                            ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------

Kavin
M                                                         104 04-SEP-26
         900
```

6. Generate Multi-Table Business Reports

Report 1: Customer Order Report

```
     SELECT c.First_Name, c.Last_Name, o.Order_ID, o.Order_Date, o.Order_Sta
tus FROM Customer c JOIN Orders o ON c.Customer_ID = o.Customer_ID;
SQL>
FIRST_NAME
--------------------------------------------------
LAST_NAME                                           ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
ORDER_STATUS
--------------------
Arun
Kumar                                                    101 01-SEP-26
Delivered

Priya
Raj                                                      102 02-SEP-26
Pending

FIRST_NAME
--------------------------------------------------
LAST_NAME                                           ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
ORDER_STATUS
--------------------

Arun
Kumar                                                    103 03-SEP-26
Shipped

Kavin
M                                                        104 04-SEP-26

FIRST_NAME
--------------------------------------------------
LAST_NAME                                           ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
ORDER_STATUS
--------------------
Delivered
```

```
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
ORDER_STATUS
--------------------
Arun
Kumar                                                   101 01-SEP-26
Delivered

Priya
Raj                                                     102 02-SEP-26
Pending

FIRST_NAME
--------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
ORDER_STATUS
--------------------

Arun
Kumar                                                   103 03-SEP-26
Shipped

Kavin
M                                                       104 04-SEP-26

FIRST_NAME
--------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
ORDER_STATUS
--------------------
Delivered

Priya
Raj                                                     105 05-SEP-26
Pending
```

Report 2: Sales Report

Display product name, quantity sold, and total revenue.

```
SQL> SELECT p.Product_Name,SUM(od.Quantity) AS Quantity_Sold, SUM(od.Quantit
y * od.Price) AS Total_Revenue FROM Product p JOIN Order_Details od ON p.Pro
duct_ID = od.Product_ID GROUP BY p.Product_ID, p.Product_Name;

no rows selected
```

Report 3: Payment Analysis Report

```
SQL> SELECT Payment_Mode,  COUNT(*) AS Total_Transactions,SUM(CASE WHEN Paym
ent_Status = 'Success' THEN 1 ELSE 0 END) AS Successful_Payments FROM Paymen
t GROUP BY Payment_Mode;

PAYMENT_MODE         TOTAL_TRANSACTIONS SUCCESSFUL_PAYMENTS
-------------------- ------------------ -------------------
Net Banking                           1                   0
Card                                  1                   0
Cash                                  1                   1
UPI                                   2                   2
```

Report 4: Customer Purchase Analysis

Find 1: Top purchasing customers

```
SQL> SELECT c.First_Name, c.Last_Name,SUM(o.Total_Amount) AS Total_Spent FRO
M Customer c JOIN Orders o ON c.Customer_ID = o.Customer_ID GROUP BY c.Custo
mer_ID, c.First_Name, c.Last_Name ORDER BY Total_Spent DESC;

FIRST_NAME
--------------------------------------------------
LAST_NAME                                          TOTAL_SPENT
-------------------------------------------------- -----------
Priya
Raj                                                       1150

Kavin
M                                                          900

Arun
Kumar                                                      800
```

Find 2: Customers with the maximum number of orders

```
SQL> SELECT c.First_Name, c.Last_Name, COUNT(o.Order_ID) AS Total_Orders FRO
M Customer c JOIN Orders o ON c.Customer_ID = o.Customer_ID GROUP BY c.Custo
mer_ID, c.First_Name, c.Last_Name ORDER BY Total_Orders DESC;

FIRST_NAME
--------------------------------------------------
LAST_NAME                                          TOTAL_ORDERS
-------------------------------------------------- ------------
Arun
Kumar                                                         2

Priya
Raj                                                           2

Kavin
M                                                             1
```

Find 3: Customers with the highest spending

```
SQL> SELECT c.First_Name, c.Last_Name,SUM(o.Total_Amount) AS Total_Spent FRO
M Customer c JOIN Orders o ON c.Customer_ID = o.Customer_ID GROUP BY c.Custo
mer_ID, c.First_Name, c.Last_Name HAVING SUM(o.Total_Amount) = (SELECT MAX(T
otal_Spent) FROM (SELECT SUM(Total_Amount) AS Total_Spent FROM Orders GROUP
BY Customer_ID));

FIRST_NAME
--------------------------------------------------
LAST_NAME                                          TOTAL_SPENT
-------------------------------------------------- -----------
Priya
Raj                                                       1150
```